# Fake News Detection System

Project Documentation

## 1. Introduction:

In today's digital era, social media has become one of the primary sources of news consumption. Along with its advantages, it has also led to the rapid spread of fake news and misinformation. Fake news can influence public opinion, create panic, and mislead users. Such content is shared not only in textual form but also through images and videos, which makes detection more challenging.

The objective of this project is to develop a **machine learning–based Fake News Detection System** that classifies news as **Fake** or **Real**. The system supports multiple input formats including text, text extracted from images using Optical Character Recognition (OCR), and transcripts generated from videos using speech-to-text techniques.

The goal of this project is not only to achieve high model accuracy but also to build a **practical, explainable, and deployable system** that works under real-world constraints.

## 2. Business Understanding

Problem Statement:
Fake news spreads very quickly on digital platforms, and manual verification of every news item is not feasible due to the large volume of data and the speed at which information spreads. Hence, there is a need for an automated system that can assist users in identifying potentially fake news content

Objectives

The key objectives of this project are:

- To classify news content as fake or real using machine learning.
- To support text, image-based, and video-based news inputs.
- To design a simple and user-friendly interface.
- To understand real-world issues such as dataset bias and deployment limitations.

## 3. Data Understanding (CRISP-DM Phase 2)

The dataset used in this project was obtained from Kaggle and consists of two files:

- **Fake.csv** – fake news articles
- **True.csv** – real news articles

The dataset primarily contains long-form news articles, most of which are related to political and global events.

**Observations**

- Fake news articles often contain sensational and emotional language.
- Real news articles usually follow a formal and neutral tone.

- The dataset is relatively balanced, which helped in fair model training.
  Understanding the nature of the dataset was important because the model learns patterns based entirely on the data it is trained on.

## 4.Data Preparation

Text Preprocessing

The following preprocessing steps were applied:

- Conversion of text to lowercase
- Removal of punctuation and special characters
- Removal of unnecessary whitespace
- Creation of a single unified content column

Why preprocessing was required:
Raw text contains noise that negatively affects feature extraction. Cleaning the text improves model learning and consistency.

### Feature Extraction using TF-IDF

TF-IDF (Term Frequency–Inverse Document Frequency) was used to convert text into numerical features.

**Why TF-IDF was chosen:**

- It assigns higher importance to meaningful words
- It reduces the influence of common words
- It is efficient and interpretable
- It works well for text classification problems such as fake news detection

### Use of Pipeline

All preprocessing, TF-IDF vectorization, and model training steps were implemented inside a Scikit-learn Pipeline.

Why Pipeline was used:

- Prevents data leakage
- Ensures identical preprocessing during training and prediction

- Simplifies deployment by saving all steps together
- Improves reproducibility and maintainability

**5. Model Selection & Training**

Several machine learning models were evaluated:

- Naive Bayes
- Support Vector Machine (SVM)
- Logistic Regression

Final Model: Logistic Regression

Why Logistic Regression was selected:

- It provided stable and consistent performance
- Faster training and inference compared to SVM
- Less prone to overfitting
- Easy to interpret
- Well-suited for TF-IDF based text classification

The model was trained using a stratified train-test split to preserve class distribution

**6. Model Evaluation & Validation**

**Evaluation Techniques Used**

- Accuracy score
- Stratified train-test split
- K-Fold cross-validation

**Results**

- Test accuracy achieved: **~98–99%**
- Cross-validation confirmed stable performance across folds

**Real-World Testing Insight**

During real-world testing, the model initially classified very short or informal text as fake.

**Reason:**
The training data mainly consisted of long news articles, so the model did not learn patterns for short casual text.

**Improvements Applied:**

- Confidence-based threshold tuning
- Rule-based handling for very short inputs

These steps reduced false positives and improved practical usability.

### 7. Deployment & Integration

A user-friendly web interface was developed using Streamlit.

Application Features

- Text-based fake news classification
- Image upload with OCR-based text extraction
- Video upload with speech-to-text processing

Why Streamlit was used:

- Simple and fast UI development
- Easy integration with machine learning pipelines
- Suitable for academic and prototype-level projects

### 8. Quality Assurance & Constraints (CRISP-ML(Q) Focus)

Technical Constraints

- OCR and video processing require system-level dependencies
- Free cloud platforms impose limitations on such dependencies

Quality Considerations

- Model predictions are probabilistic, not absolute truths
- System performance depends on input quality (image clarity, audio quality)
- Dataset bias affects generalization to casual or short text

As a result, full multimodal functionality is demonstrated through local deployment, while the source code and documentation are maintained on GitHub.

## 9. Limitations

- The model performs best on news-style text and may not generalize well to informal language.
- The system does not verify factual correctness using external sources.
- OCR and speech-to-text accuracy depend on input quality.
- Only English-language content is supported.
- Cloud deployment is limited due to system-level dependency restrictions.

## 10. Future Improvements

- Integration of transformer-based models such as BERT
- Multilingual fake news detection
- Fact-checking using trusted APIs
- Improved OCR and speech recognition models
- Container-based cloud deployment using Docker

## 11. Conclusion

This project demonstrates the development of an end-to-end fake news detection system following the **CRISP-ML(Q)** approach, focusing not only on model accuracy but also on data quality, deployment constraints, and real-world reliability. The project highlights the importance of structured ML workflows, model validation, and quality considerations in building practical machine learning systems.